In [ ]:
```
# Import the packages
# read the data
# divide into categorical and numerical
# create a frequncy table using value counts
# and save at a python folder
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path=r'C:\Users\omkar\OneDrive\Documents\Gen_AI\Data_files\Visadataset.csv'
visa_df=pd.read_csv(file_path)

cat=visa_df.select_dtypes(include='object').columns
num=visa_df.select_dtypes(exclude='object').columns
```

In [3]:
```
visa_df['continent'].value_counts()
keys=visa_df['continent'].value_counts().keys()
values=visa_df['continent'].value_counts().values
df=pd.DataFrame(zip(keys,values),
                columns=['Continent','Number of Applicants'])
df.to_csv('continent_df.csv')
```

In [5]:
```
df
```

Out[5]:

	Continent	Number of Applicants
0	Asia	16861
1	Europe	3732
2	North America	3292
3	South America	852
4	Africa	551
5	Oceania	192

create a new folder

- I want to save in a folder that folder should avaialble in python location

In [15]:
```
import os
new_folder_name='frequency_tables'
try:
    os.makedirs(new_folder_name)
except Exception as e:
    print(e)
```

```
[WinError 183] Cannot create a file when that file already exists: 'frequency_tab
les'
```

In [ ]:
```
C:\Users\omkar\OneDrive\Documents\Data science\Naresh IT\Naresh IT\Data science\
C:\Users\omkar\OneDrive\Documents\Data science\Naresh IT\Naresh IT\Data science\
```

In [19]:
```
visa_df['continent'].value_counts()
keys=visa_df['continent'].value_counts().keys()
values=visa_df['continent'].value_counts().values
df=pd.DataFrame(zip(keys,values),
             columns=['Continent','Number of Applicants'])
df.to_csv(r'C:\Users\omkar\OneDrive\Documents\Data science\Naresh IT\Naresh IT\D
```

In [ ]:
```
'C:\Users\omkar\OneDrive\Documents\Data science\Naresh IT\Naresh IT\Data science
frequency_tables\
continent_df.csv'
```

In [25]:
```
os.getcwd()+'\\'+new_folder_name
```

Out[25]:
```
'C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Naresh IT\\Dat
a science\\Batches\\Batch-15_Apr\\EDA\\frequency_tables'
```

In [29]:
```
cwd=os.getcwd()  # python location
new_dir=os.path.join(cwd,new_folder_name) # folder location
file_name='continent.csv'
new_path=os.path.join(new_dir,file_name) # path
new_path
```

Out[29]:
```
'C:\\Users\\omkar\\OneDrive\\Documents\\Data science\\Naresh IT\\Naresh IT\\Dat
a science\\Batches\\Batch-15_Apr\\EDA\\frequency_tables\\continent.csv'
```

In [31]:
```
df.to_csv(new_path)
```

In [ ]:
```
import os
new_folder_name='frequency_tables'
try:
    os.makedirs(new_folder_name)
except Exception as e:
    print(e)

cwd=os.getcwd()
new_dir=os.path.join(cwd,new_folder_name)
file_name='continent.csv'
new_path=os.path.join(new_dir,file_name) # path
new_path
```

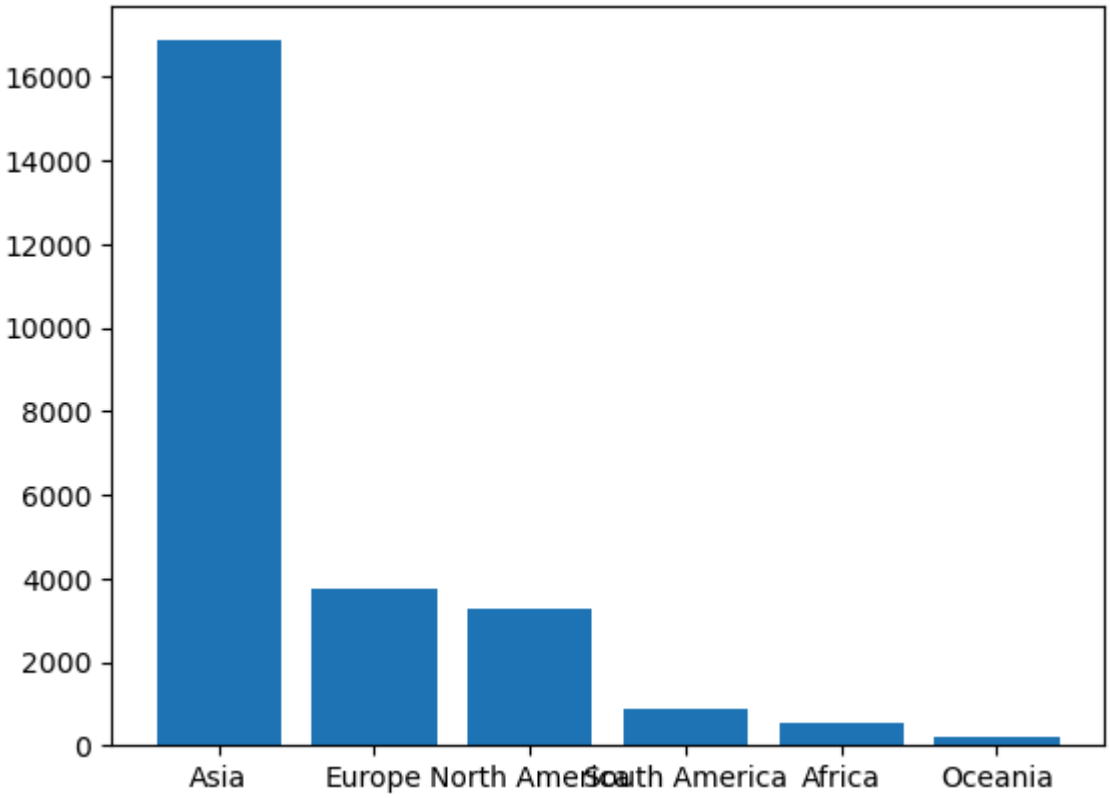
In [35]:
```
cwd=os.getcwd()
new_dir=os.path.join(cwd,new_folder_name)
#####################################################################
visa_df['continent'].value_counts()
keys=visa_df['continent'].value_counts().keys()
values=visa_df['continent'].value_counts().values
df=pd.DataFrame(zip(keys,values),
             columns=['Count','Number of Applicants'])
#############################################################
file_name='continent.csv'
new_path=os.path.join(new_dir,file_name) # path
df.to_csv(new_path)
```

In [39]:
```
cwd=os.getcwd()
new_dir=os.path.join(cwd,new_folder_name)
#####################################################################
for i in cat[1:]:
    visa_df[i].value_counts()
```

```
    keys=visa_df[i].value_counts().keys()
    values=visa_df[i].value_counts().values
    df=pd.DataFrame(zip(keys,values),
                columns=['Count','Number of Applicants'])
    ###############################################################
    file_name=f'{i}.csv'
    new_path=os.path.join(new_dir,file_name) # path
    df.to_csv(new_path)
```

uploading bar charts

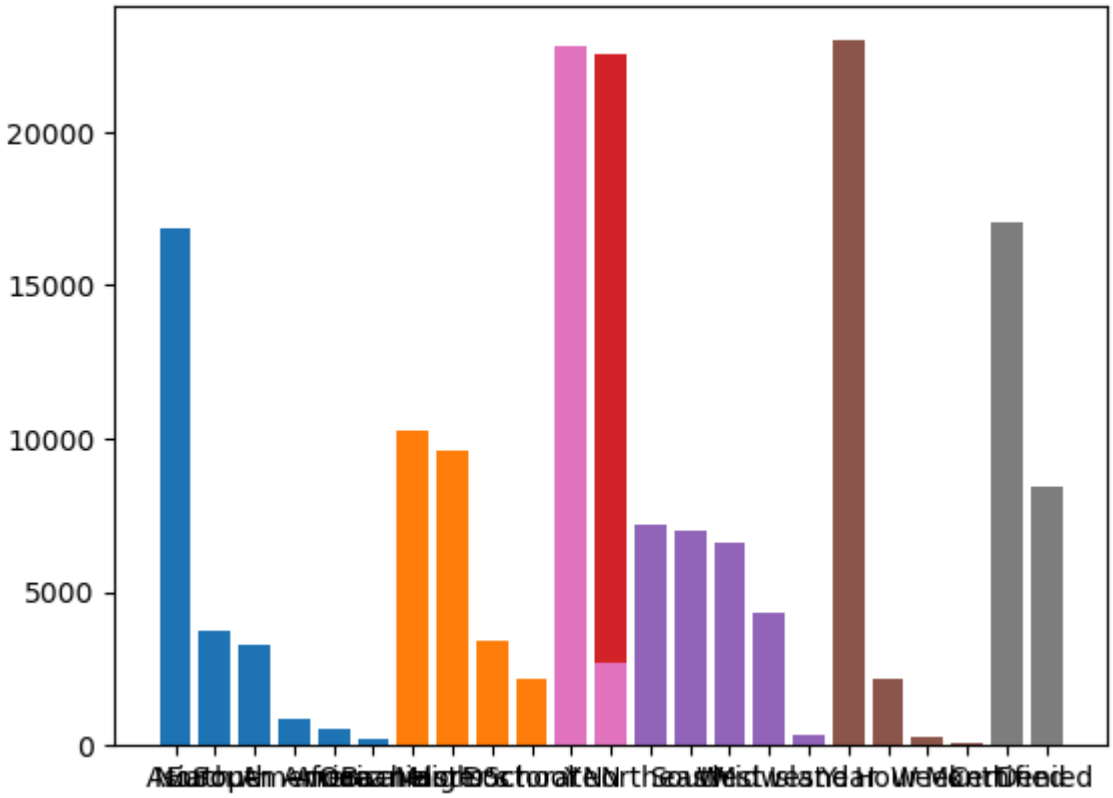
In [8]:
```
keys=visa_df['continent'].value_counts().keys()
values=visa_df['continent'].value_counts().values
plt.bar(keys,values)
plt.savefig('continent_bar.jpg')
```



In [12]:
```
import os
folder_name='bar_charts'
try:
    os.makedirs(folder_name)
except Exception as e:
    print(e)

cwd=os.getcwd()
new_dir=os.path.join(cwd,folder_name)

for i in cat[1:]:
    keys=visa_df[i].value_counts().keys()
    values=visa_df[i].value_counts().values
    file_name=f'{i}_bar.jpg'
    path=os.path.join(new_dir,file_name)
    plt.bar(keys,values)
    plt.savefig(path)
```



In [ ]: